

Mohak Kapoor

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SUMMARY

ML Engineer and a final-year student with a knack for building end-to-end systems. Built a meta-labeling framework on Indian market data achieving Sharpe 1.48 on GoldBees OOS; engineered a two-stage hybrid IDS on 2.8M rows with LightGBM at 99% accuracy, deployed live on Raspberry Pi 5 via FastAPI and Cloudflare Tunnel. Interested in financial and production ML.

PROJECTS

METR — Market Exposure Timing Research — Python, Polars, SQL, XGBoost, Feature Engineering *Mar 2026 – Apr 2026*

- Constructed a **two-layer meta-labeling framework** on 3,440+ days of multi-asset Indian market data (Nifty 50, GoldBees, USD/INR), using Polars and **SQL-based validation queries** with signal-conditional masking and 30 cross-asset features for **time-series classification** and trade entry quality filtering via XGBoost.
- Sharpe 1.48, Calmar 1.87, 66.1% OOS win rate (2024–2025)** significant at $p < 0.05$ across three independent tests, validated against **10,000 Monte Carlo simulations**; null results on Nifty and USD/INR consistent with market efficiency theory.
- Preserved 81% historical memory with full stationarity** via Fractional Differentiation ($d = 0.45$) and **Triple Barrier volatility-adaptive labeling**; implemented exploratory data analysis and SHAP feature importance analysis which identified Vol Efficiency and VIX Momentum as primary alpha drivers consistent with GoldBees' rupee-dollar exposure.

Network IDS Detection Engine — Python, LightGBM, XGBoost, PyTorch, scikit-learn, FastAPI, Docker *Jan 2026 – May 2026*

- Engineered a **two-stage hybrid IDS** on 2.8M rows of CICIDS2017. Unsupervised anomaly detection flags candidates for supervised classification, mirroring production platforms like Snort and Darktrace; built **end-to-end data pipeline** handling ingestion, EDA, preprocessing, and temporal train/test splitting to prevent distribution leakage across unseen attack types.
- LightGBM 99% accuracy (F1: 0.99)** across four benchmarked architectures (XGBoost 97%, PyTorch FFNN 98%, SVM 96.7%); Denoising Autoencoder (AUC: 0.7801) outperformed Isolation Forest on zero-day detection, with IncrementalPCA **cutting feature space 51%** (98.97% variance retained).
- Deployed a publicly accessible real-time threat monitoring system** by shipping a Dockerized FastAPI inference engine on Raspberry Pi 5 for live model deployment, with WebSocket streaming dashboard with latency metrics, and Bearer token auth via Cloudflare Tunnel.

CaptchaOCR — CAPTCHA Recognition System — Python, PyTorch, CNN, BiLSTM, Computer Vision *Dec 2025*

- Trained a CAPTCHA recognition system using a **CNN + BiLSTM (CRNN) architecture with CTC loss**, achieving **96%+ character-level accuracy** on distorted, variable-length sequences via synthetic training corpus generation and noise-augmentation preprocessing.
- Enabled reproducible benchmarking and open community validation** by publishing to **Hugging Face Spaces** for public model deployment and inference, extending impact beyond offline experimentation.

EXPERIENCE

HumanizeIQ

AI Intern

Jun 2025 – Oct 2025

Hazlet, New Jersey (Remote)

- Consolidated 3 manual pipeline steps into a single API call** by engineering production **MCP servers on Cloudflare Workers** for client-facing async image generation and multimodal document workflows via persistent R2 storage.
- Cut analyst turnaround from 2 hours to under 5 minutes** by building **LLM pipelines (GPT-4, Gemini)** for automated call transcription and structured report generation, eliminating manual post-call processing.
- Supported 100% of the platform's background processing load** by designing **REST APIs** for async job orchestration enabling reliable long-running LLM inference within chat workflows; led daily Scrum stand-ups in an Agile environment.

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++
ML & Modeling	Supervised Learning, SVMs, LightGBM, XGBoost, CNN, BiLSTM, CRNN, Time-Series Forecasting, Anomaly Detection, Statistical Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning
LLMs & GenAI	LLM Integration (GPT-4, Gemini), Multimodal Pipelines, MCP, RAG
MLOps & Deployment	FastAPI, REST APIs, Docker, CI/CD (GitHub Actions), MLflow, Experiment Tracking, Model Monitoring, Cloudflare Workers, Cloudflare Tunnel, Edge Deployment (Raspberry Pi 5)
Frameworks & Data	TensorFlow, PyTorch, scikit-learn, Django, Hugging Face, Pandas, NumPy, Polars, PostgreSQL, SQL, Jupyter, Git, Linux
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, Computer Networks

EDUCATION

Jaypee Institute of Information Technology

B.Tech in Electronics and Communication Engineering

Noida, India

2022-2026

- Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Multivariate & Differential Calculus, Control Systems